

Ray Shang

Curriculum Vitae

✉ rshxng@gmail.com

📄 <https://ray-shang.github.io/>

Education

- 2025-Present **PhD Candidate**, *University of Cambridge*, Department of Pure Mathematics and Statistics, Cambridge, UK.
- 2020-2024 **BA, Mathematics**, *Harvard College*, Cambridge, MA, GPA: 3.988/4.000.

Papers

- 2024 **Ray Shang**, *Interpolation for degree 2 Veroneses of odd dimension*, arXiv:2411.16672.
- 2024 **Ray Shang**, *Slope semistability for Veronese normal bundles*, arXiv:2411.16664.

Research Experience

Hausdorff Center for Mathematics

- September 2024 - Present **K3 Surfaces**.
- 2024 - Investigated K3 surfaces and moduli of Gieseker (semi)stable sheaves.
- Advisor: **Daniel Huybrechts**, *Professor, Department of Mathematics*, University of Bonn
- Harvard College**
- May 2023 - **Semistability and Interpolation for Veronese normal bundles**.
- May 2024 Proved slope semistability for all Veronese normal bundles and proved interpolation for degree 2 Veroneses of odd dimension.
- Advisors: **Joe Harris**, *Professor, Department of Mathematics*, Harvard College, **Anand Patel**, *Professor, Department of Mathematics*, Oklahoma State University
- University of Chicago**
- May 2022 - **UChicago Research Experience for Undergraduates**.
- December 2022 Attended the University of Chicago's research experience for undergraduates, funded by the National Science Foundation. Investigated the relation between complex orientations of generalized cohomology theories to Chern classes and formal group laws, culminating in an expository article.
- Advisor: **Peter May**, *Professor, Department of Mathematics*, University of Chicago

Fellowships & Awards

- 2024 Recipient of DAAD **Research Grant - One Year Grants for Doctoral Candidates**. Research grant awarded by Germany's premier funding institution for those with "excellent qualifications and are already taking a doctoral degree... or in exceptional cases a bachelor's degree." Nominated by Harvard University as a college senior among graduate students.
- 2023 Recipient of Harvard **PRISE Fellowship**. Harvard college's summer research fellowship; sole mathematics project selected that summer.

Teaching Assistantship

Harvard College

- Spring, 2022 : **Math 131: Introduction to Topology**.

Fall, 2022 : **Math 231: Graduate Algebraic Topology.**
Spring, 2023 : **Math 113: Introduction to Complex Analysis.**
Fall, 2023 : **Math 112: Introduction to Real Analysis.**
Spring, 2024 : **Math 121: Linear Algebra and Applications.**

Extracurriculars / Outreach

2020-2022 **Harvard-MIT Math Tournament**, The premier high school mathematics tournament in the United States; helped coordinate logistics for the tournament; emcee for educational speaker event.

2020 **Ross Mathematics Program Junior Counselor**, Guided first-years through Number Theory proofs; taught real analysis and group theory seminars.

2018-2020 **Junior High Mathematics Tournament Director**, Directed a junior high mathematics tournament; 700+ students, 20+ schools.